

THE EMERGENT BELT

Jingzhang Hyper Line

When the constraint cannot be changed, what can design still do?

Turn constraints into the conditions for emergence — measure the disease (28 computations), prescribe (stitch + entangle + unlock), prove it works (3 gates PASS).

Emergence Index
0.795 → 0.962

Percolation
0.840 → 0.904

Syntax integration
+36%

Jobs-housing entropy
+5.3%

Net λ (diagnostic)
24.76%

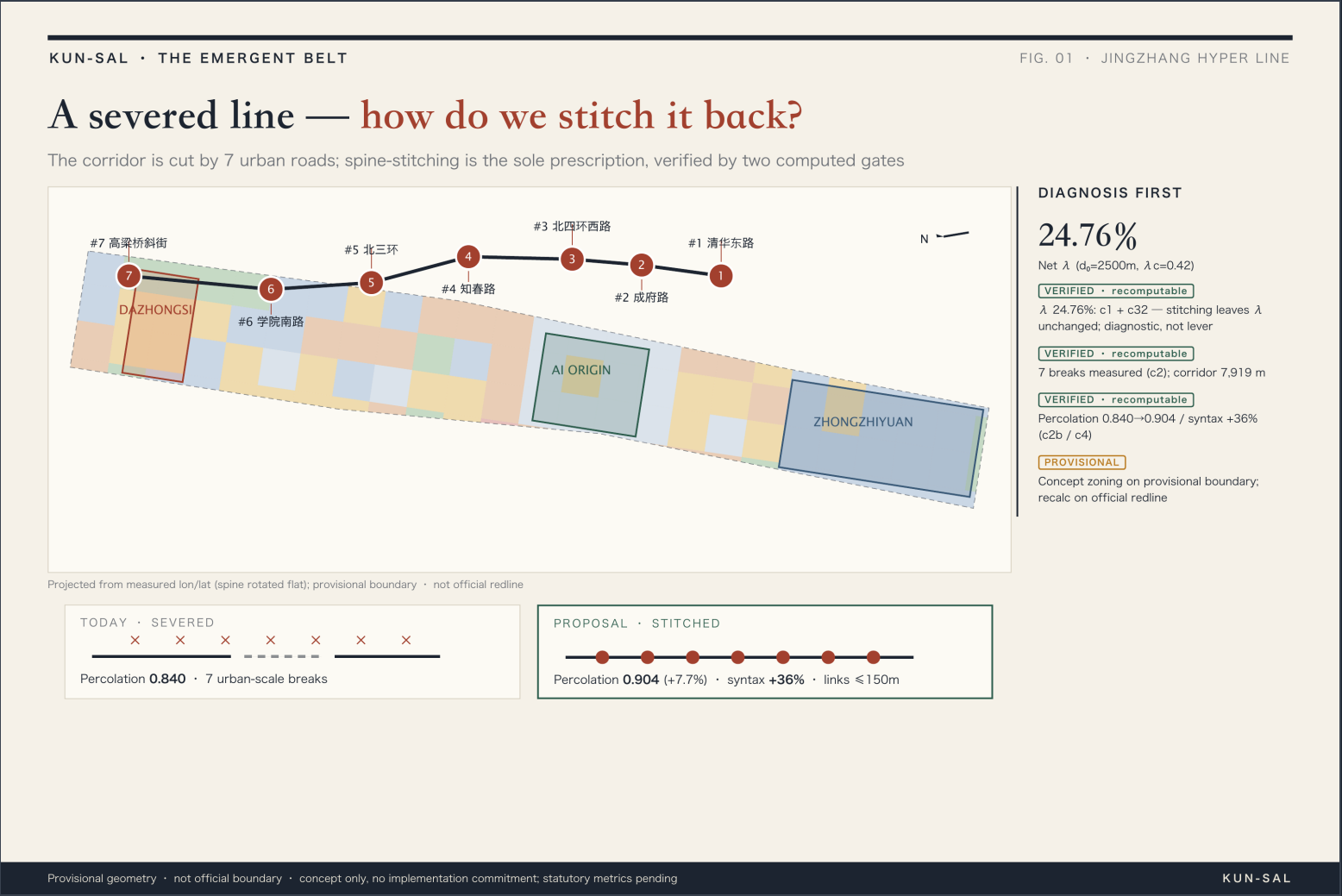


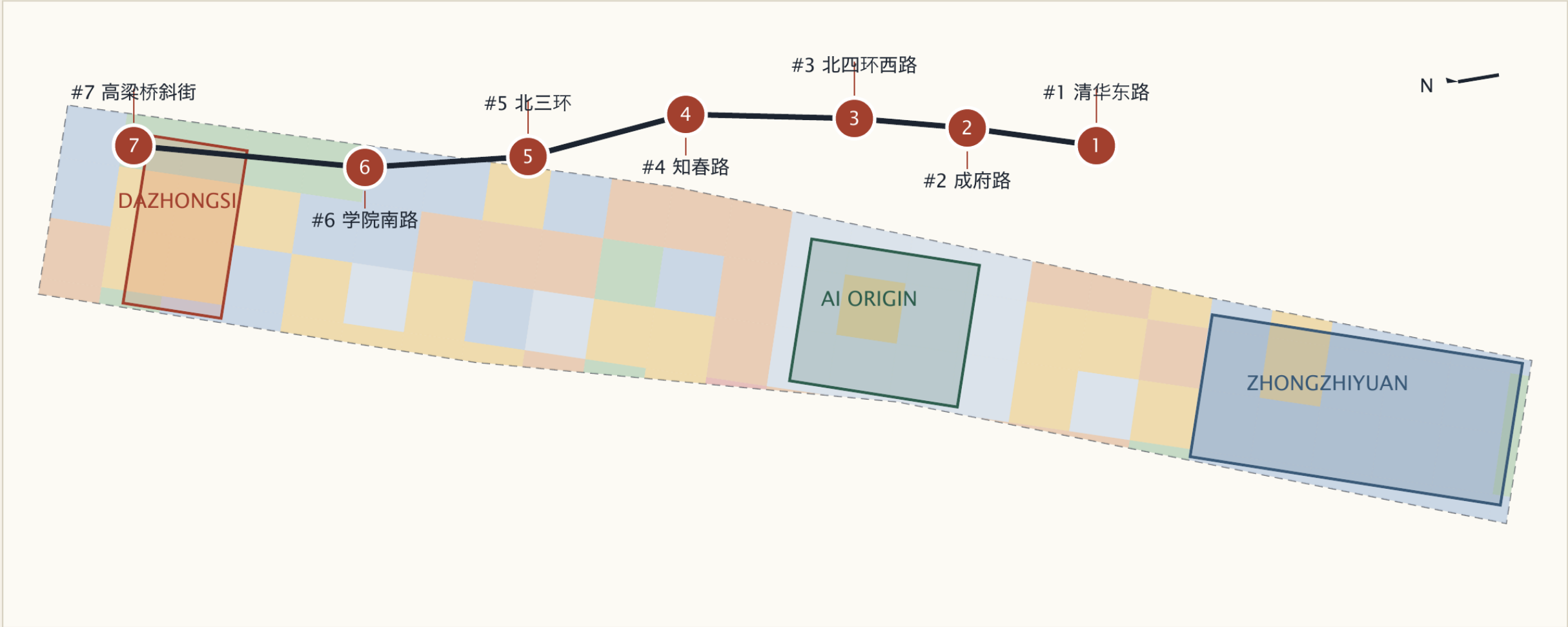
FIG.01 · severance diagnosis & spine stitching (measured geometry)

7 breaks measured

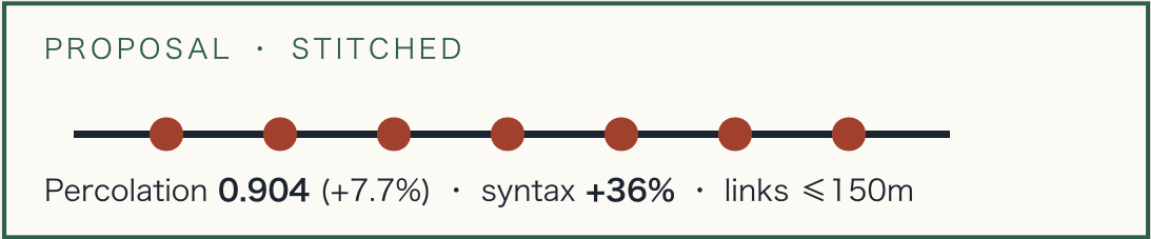
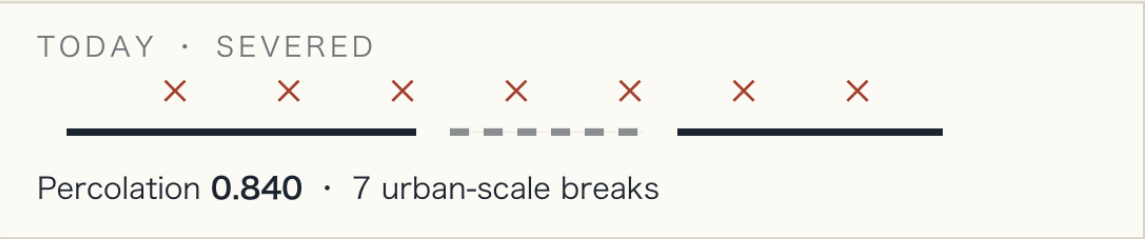
7 处 • 7,919m

A severed line — how do we stitch it back?

The corridor is cut by 7 urban roads; spine-stitching is the sole prescription, verified by two computed gates



Projected from measured lon/lat (spine rotated flat); provisional boundary · not official redline



DIAGNOSIS FIRST

24.76%

Net λ ($d_0=2500m$, $\lambda c=0.42$)

VERIFIED · recomputable

λ 24.76%: $c1 + c32$ — stitching leaves λ unchanged; diagnostic, not lever

VERIFIED · recomputable

7 breaks measured (c2); corridor 7,919 m

VERIFIED · recomputable

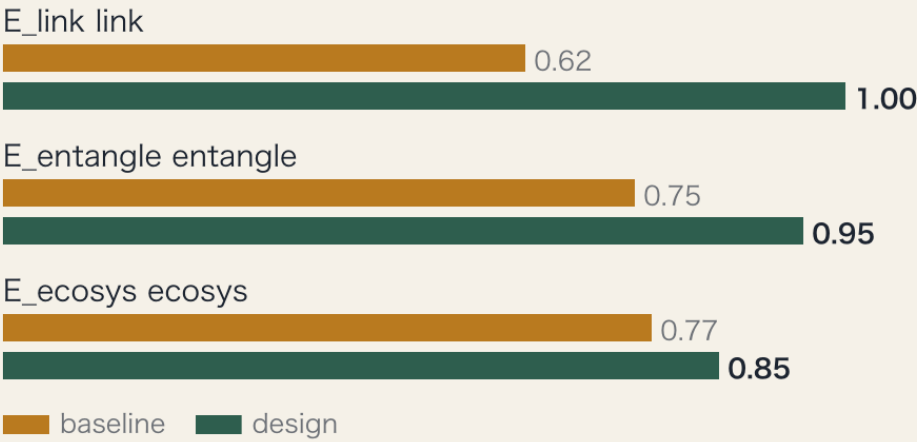
Percolation 0.840→0.904 / syntax +36% (c2b / c4)

PROVISIONAL

Concept zoning on provisional boundary; recalc on official redline

EI 0.795 → 0.962: a recomputable rise in urban health

All three gates PASS: percolation 0.904 · syntax +36% · entropy +5.3%



G1 · PERCOLATION

0.904

gate ≥0.90 · PASS

G2 · SYNTAX

+36%

gate ≥30% · PASS

G3 · ENTROPY

+5.3%

gate ≥4% · PASS

RECOMPUTABLE · HONEST LEDGER

VERIFIED · recomputable

c1 λ · c2 percolation · c4 syntax · c28 entropy · c34 ecosystem — formulas registered

VERIFIED · recomputable

K008: λ unchanged by stitching; gates use design-impact metrics

PROVISIONAL

E_ecosys 0.85 concept target (–20% closure pending)

UNKNOWN

Redline/capacity/elevation: UNKNOWN ≠ 0; no offsetting

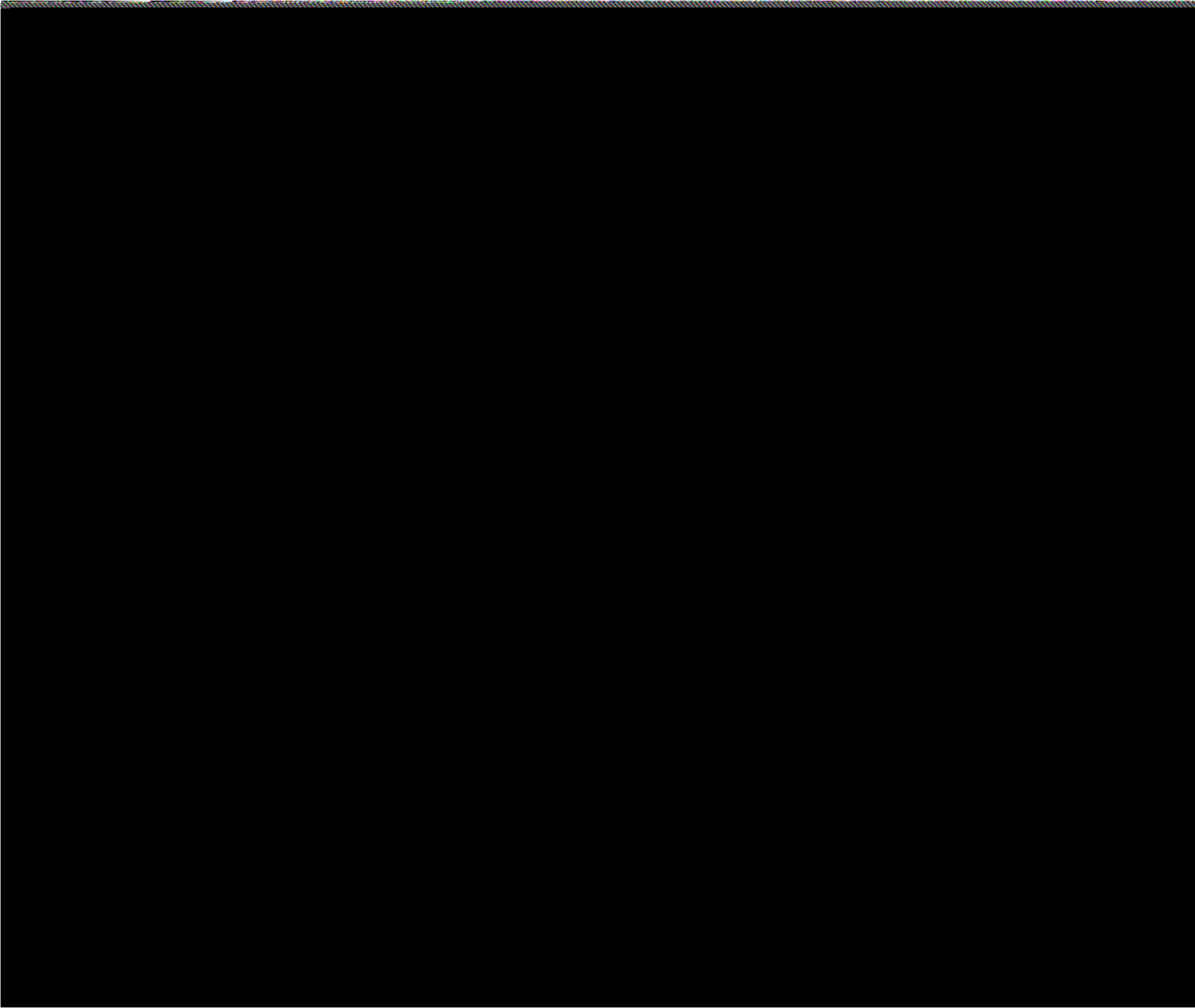
01

Diagnosis: what blocks this belt

Five scientists' consultation · Space syntax · LUTI (jobs-housing dynamics) · POI dynamics · Boundary correction

① Spatial interaction λ: stuck subcritical

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- Only one of ten core pairs crosses the line (Origin × Xiaoyuehe 0.455)
- Zhongzhiyuan×Dazhongsi λ=0.033: adjacent on the map, unreachable in ten minutes
- Not without vitality — vitality chopped apart by seven roads
- The leverage point is the seven gaps; no even spreading

Basis: $\lambda = \exp(-d/2500m)$ proxy; recalculation on official data



② POI check-up: demand recognized by 25,476 points

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▪ 25,476 POIs across the belt (2026-08-15 snapshot)

▪ Life services over-supplied; innovation content under-supplied

▪ Finance -63.0% / research -45.9% / Origin medical 2.88/km², the lowest

▪ Dazhongsi 511/km², the densest; jobs/life 0.344: life dense, work thin

Data: Amap aggregate indicators; no raw points published

25,476	-63.0%	-45.9%
POI	finance	research

③ Space syntax: configuration shapes movement

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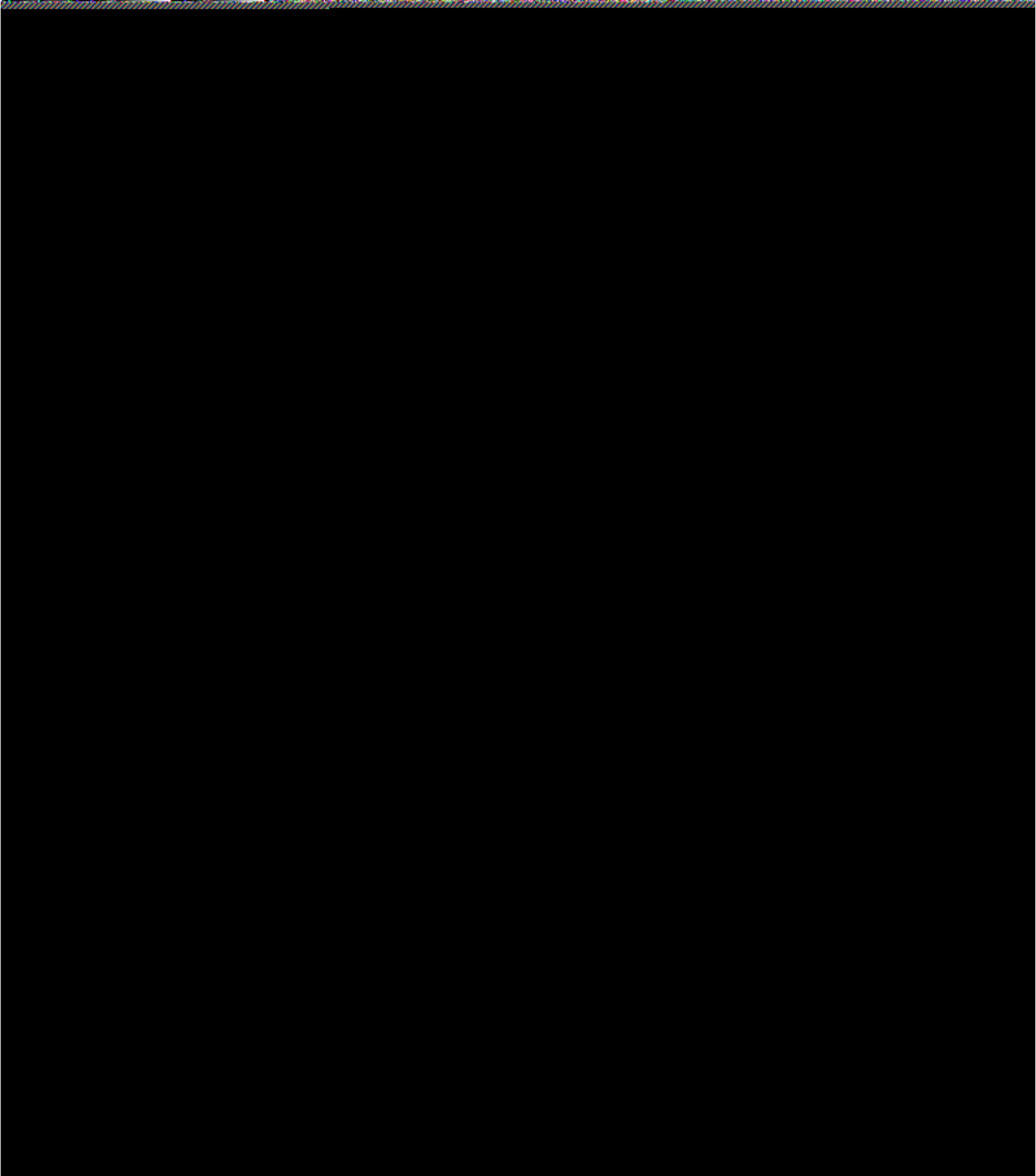
- Xueyuanqiao 97.5%: the belt's mouth at the N4th Ring × Xueyuanlu
- Tsinghuayuan 93.5%: a 1909 siting still the structural core
- Dazhongsi 20.2%: syntactic periphery, vitality must come from function
- After stitching: Dazhongsi +70%, Beijing North +49%, Wudaokou +43%

5,720-node network, computed; low intelligibility is normal for organic fabric



⑤ Deep LUTi: LIVE-WORK MOIRÉ — dissolve the commute

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▪ Disruptive concept: 45 underdeveloped cells rebuilt as 1.0 ± 0.2 live-work interlocks

▪ Commute self-containment 78.0% → 89.5% (+11.5pp, computed)

▪ Mismatch quadrants 45/4/42: where it's wrong → after stitching → what remains

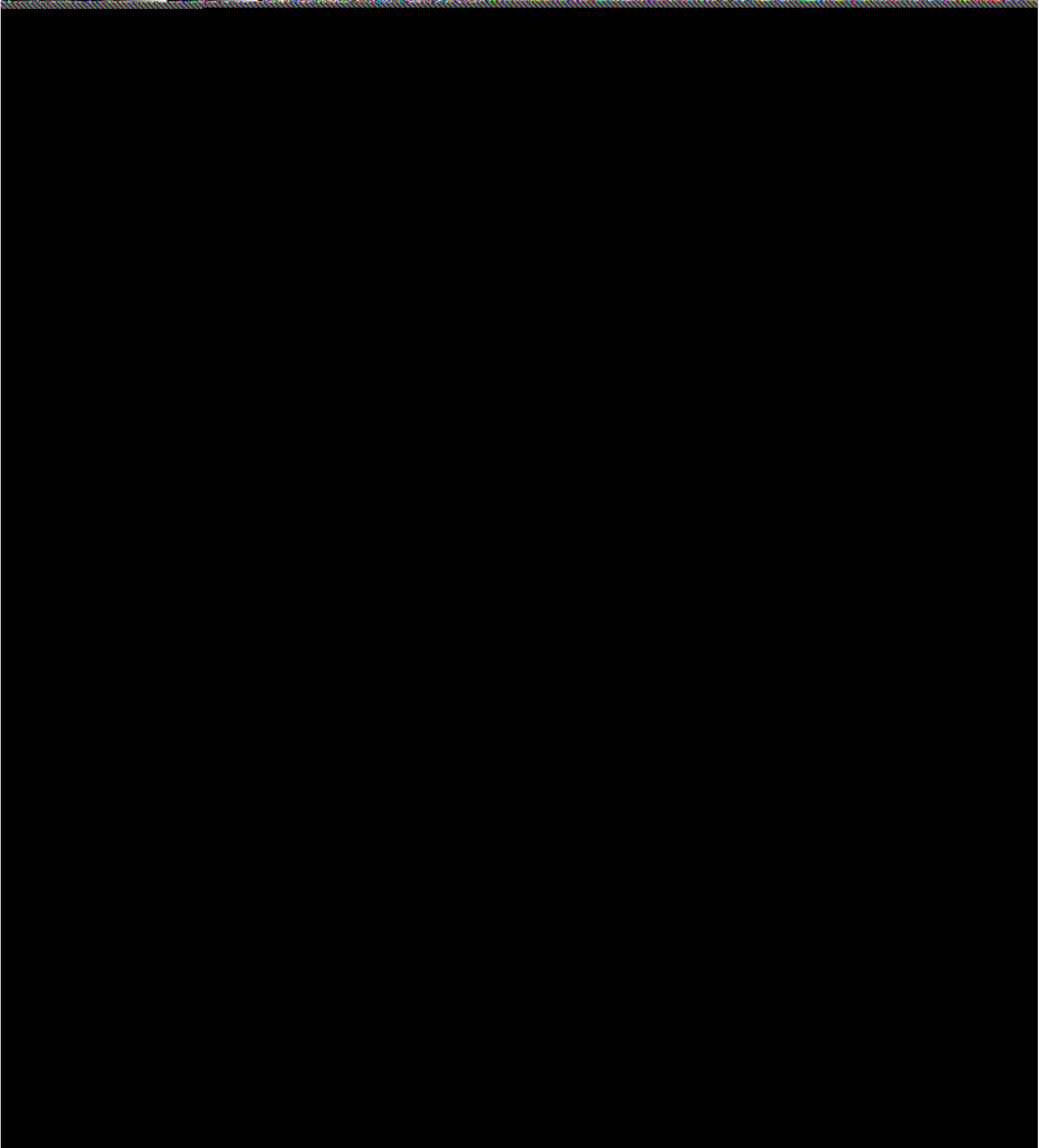
▪ Not a city with fewer commutes — a city that does not need commuting

POI proxies; recalculation on official population and job data



⑥ Three-layer LUTI scenarios: concentration lengthens, entanglement dissolves

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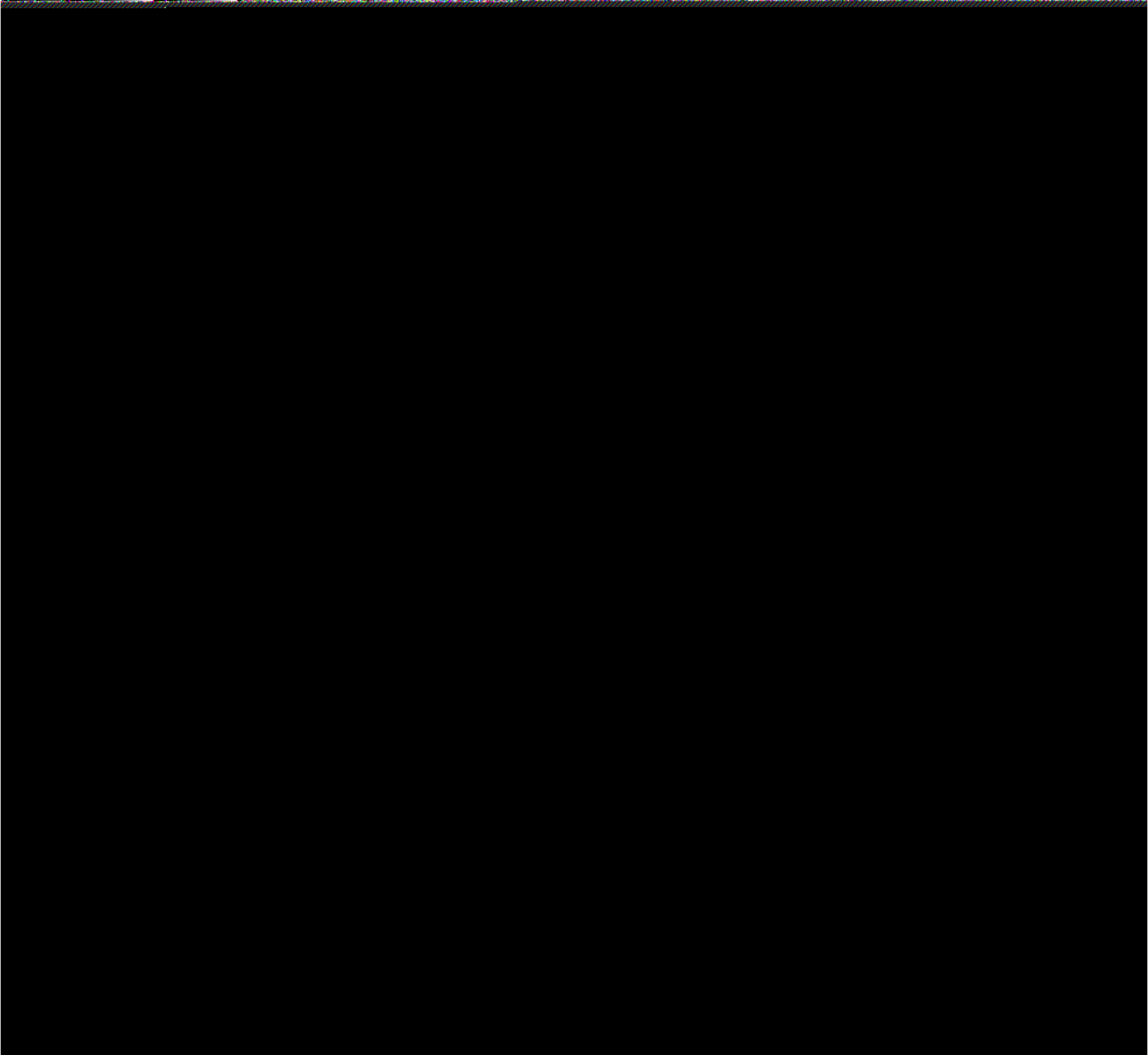
- The reveal: Wudaokou 141/km² is the real job core; Xuezhiyuan ratio 2.78, most imbalanced
- Three scenarios (gravity OD): concentration +3.9% distance, walking self-containment 7.0%→3.5%
- Entangled: entropy S +5.3% (more mixed); rules = paired infill + mixed towers + S in the annual check-up
- 1909 the switch killed the gradient, 2026 entanglement kills the commuting tide

Gravity OD + POI proxies; recalculation on official OD; floorspace ledger = the brain's first new database



⑦ Boundary correction: fixing the brief's rough boundary

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- The brief's PROV-KEY-003 rough boundary sits near Beijing North (39.944–39.950)
- Its own basis table names Dazhongsi Station (39.9653) — ~2 km away
- missing-data.md: precise polygons are missing
- Corrected by name semantics, assumption A-BOUNDARY-003, full recalculation on official data

Integrity first: a data mismatch must be disclosed, never quietly fixed

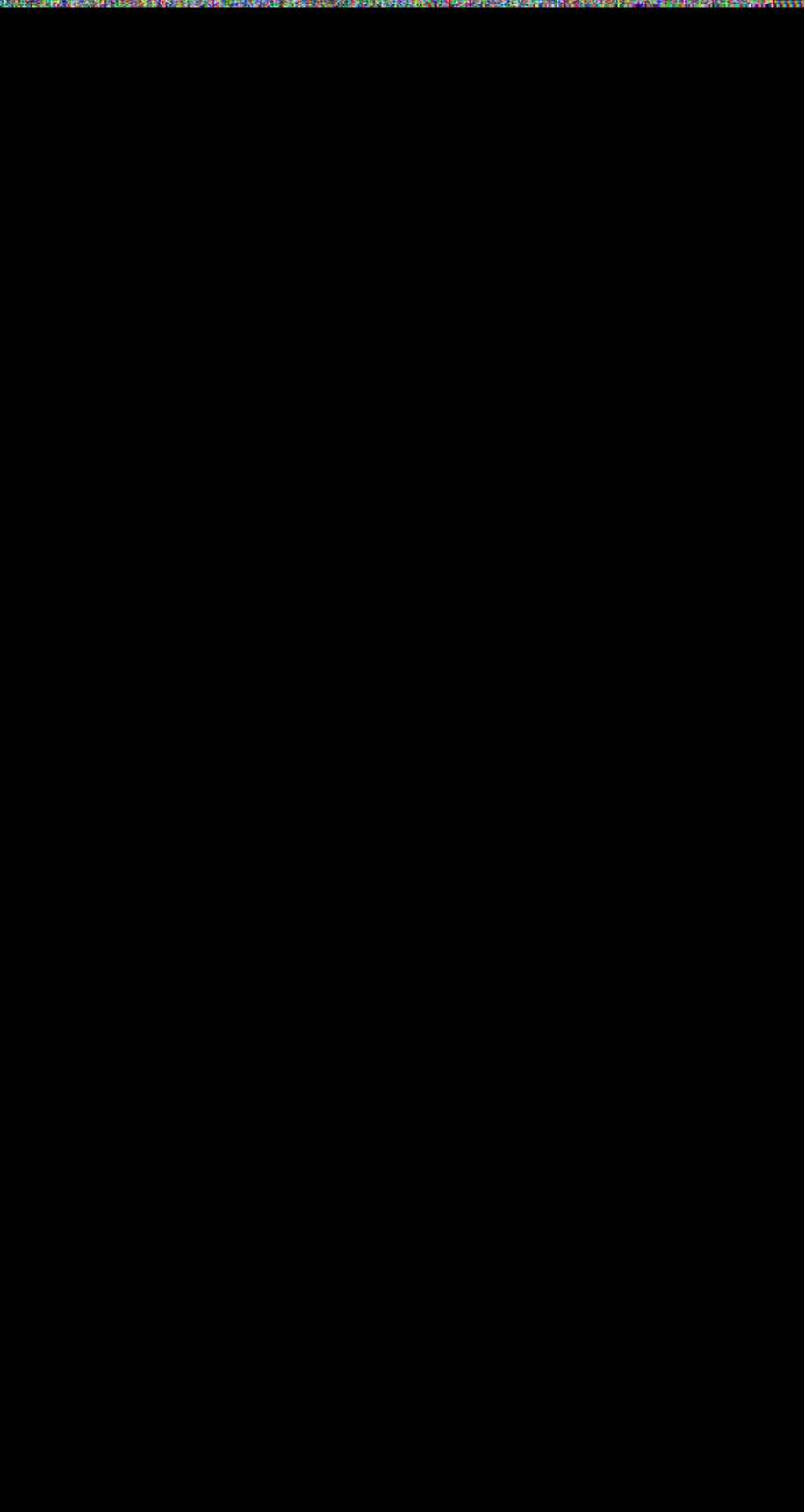
02

Spatial proposal: one line, three folds, two wings, seven nodes

Master plan · Key-area plans & sections · Character & height · Dazhongsi quadrants

⑤ Master plan (real fabric + concept overlay)

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▪ Light grey = OSM building fabric; white-cased dark = real Jing-Zhang railway

▪ Dark green = Hyper Line spine; red dots = seven stitches; stars = three landmarks

▪ Three coloured frames = three gates: STACK / ORIGIN / FRONT

▪ The base is the real city; only the overlay is design

Base: OSM (© OpenStreetMap contributors)

⑥ Zhongzhiyuan · STACK (192.1 ha)

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▪ Qinghe full-stack test band: bounded tests in garden blocks

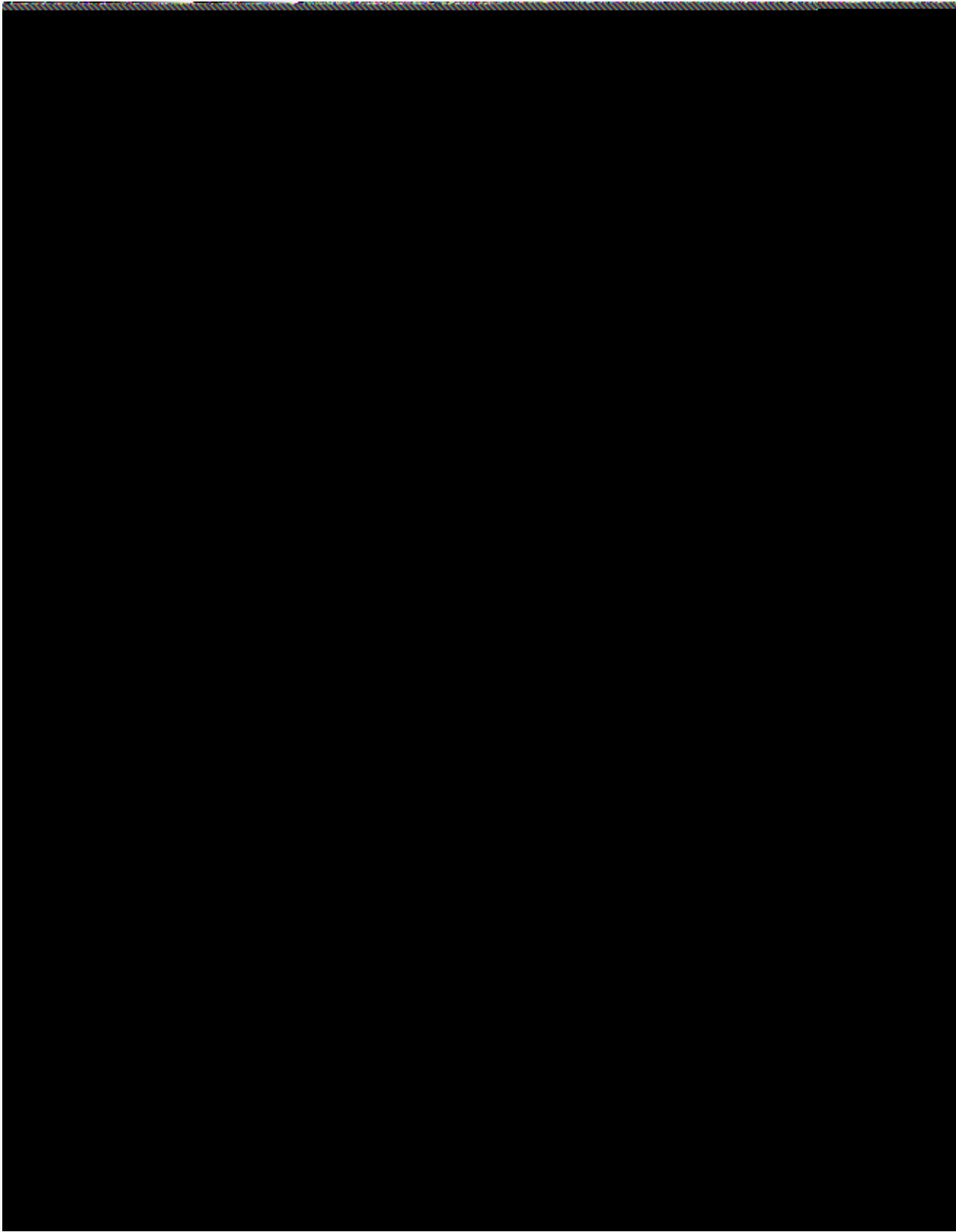
▪ Weakest: culture-tourism/education/finance → reasons to stay

▪ Section A-A': 3.2m footpath + 2.8m cycle band + test-equipment slots

▪ Long term (5–10y): garden-district renewal

⑦ Origin · ORIGIN (104.3 ha)

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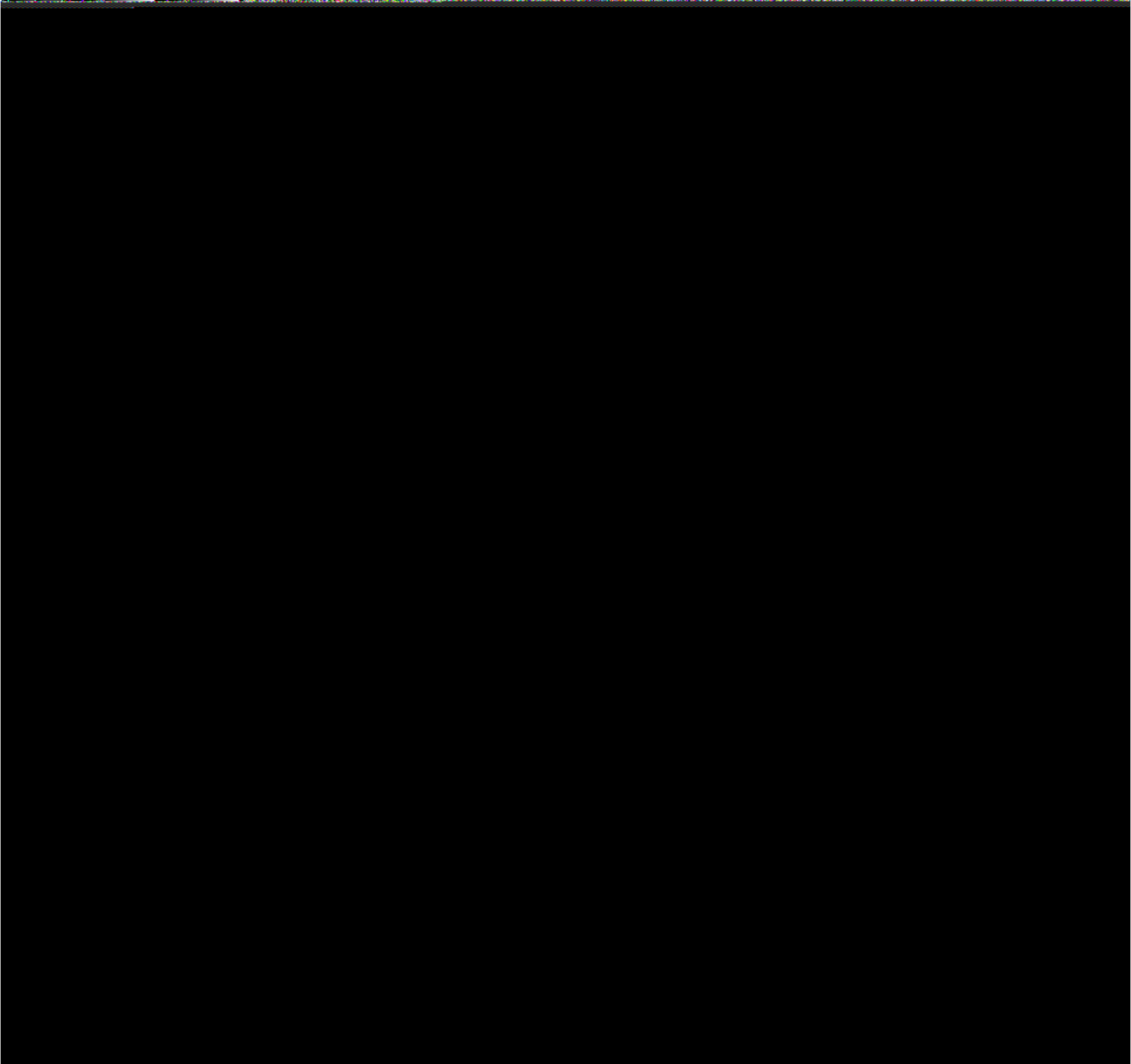


- Tsinghuayuan · Origin Plaza + time-axis name field: integration 93.5%
- Wudaokou · Phase Plaza: the λ gauge — the city's thermometer
- Medical 2.88/km², the lowest → infill healthcare/apartments/services
- Mid term (3–5y): near-campus innovation street

104.3 ha	93.5%	2.88/km²
direction	structure origin	material density

⑧ Dazhongsi · FRONT (72.0 ha)

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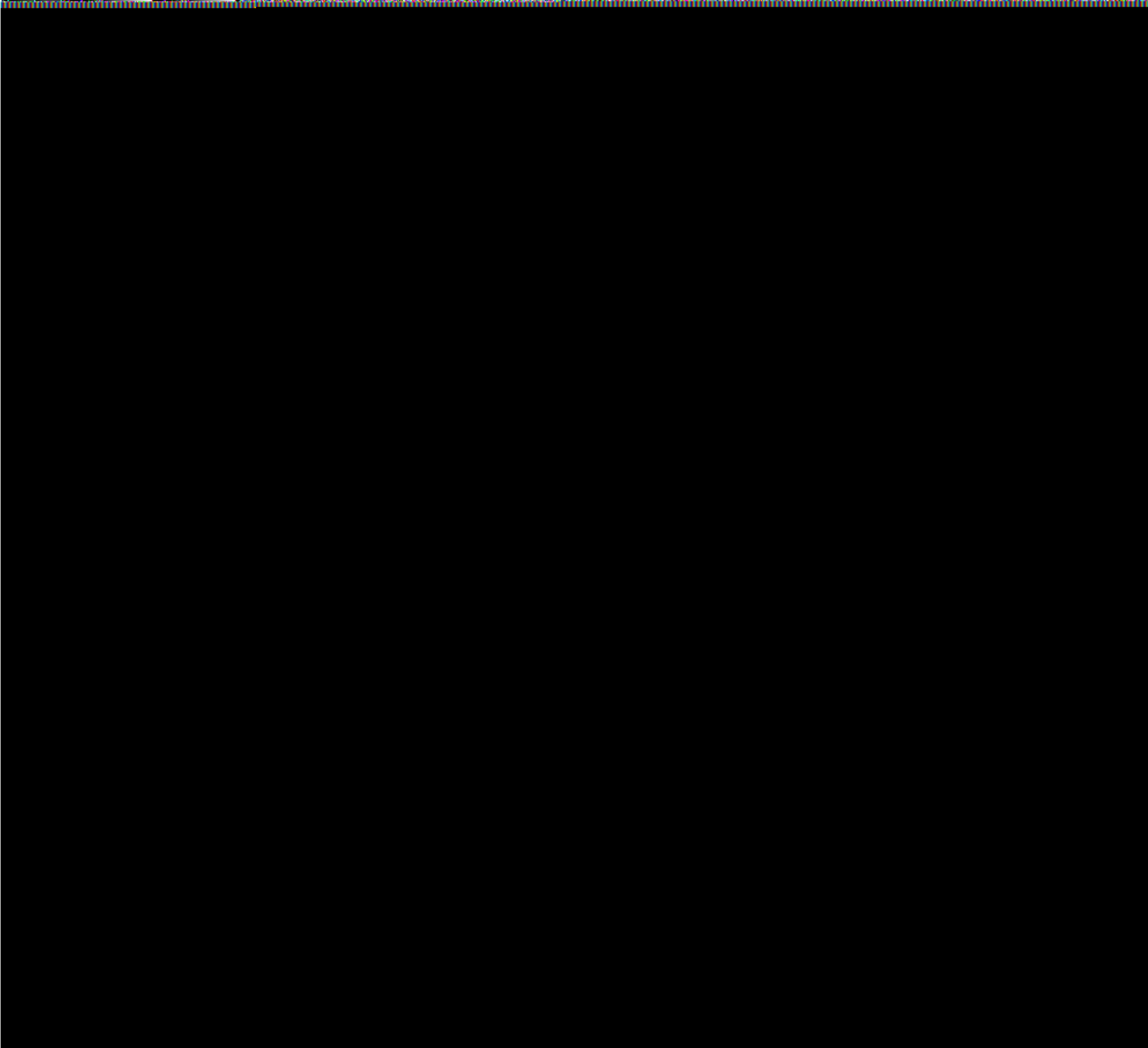


- Syntactic periphery 20.2%, yet the densest POIs at 511/km²
- Jobs/life 0.344: life-service-led → infill finance/culture-tourism/research
- Light intervention, heavy interface: stitch four quadrants; nights for classes & merchants
- Near term (1–3y): spine stitching + Dazhongsi interface



⑨ Dazhongsi four-quadrant stitching (enlarged)

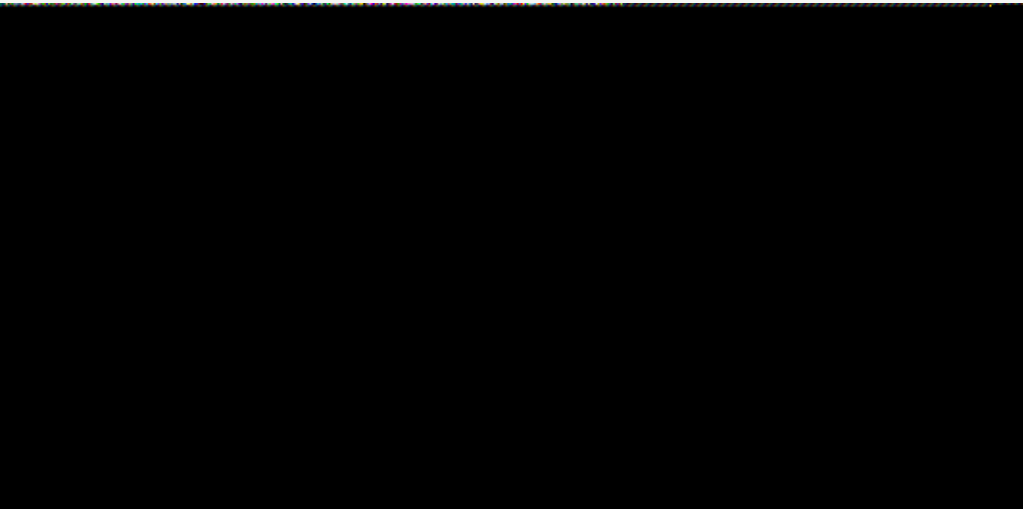
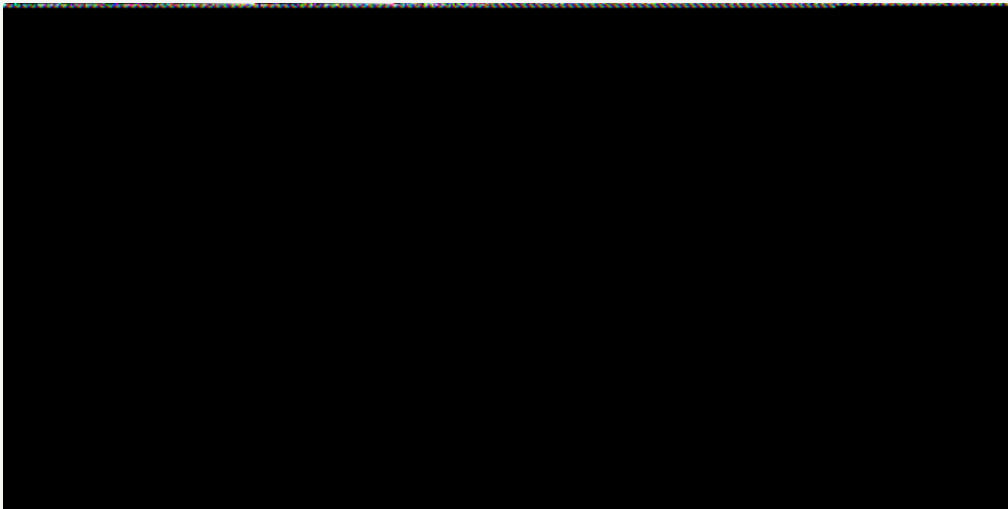
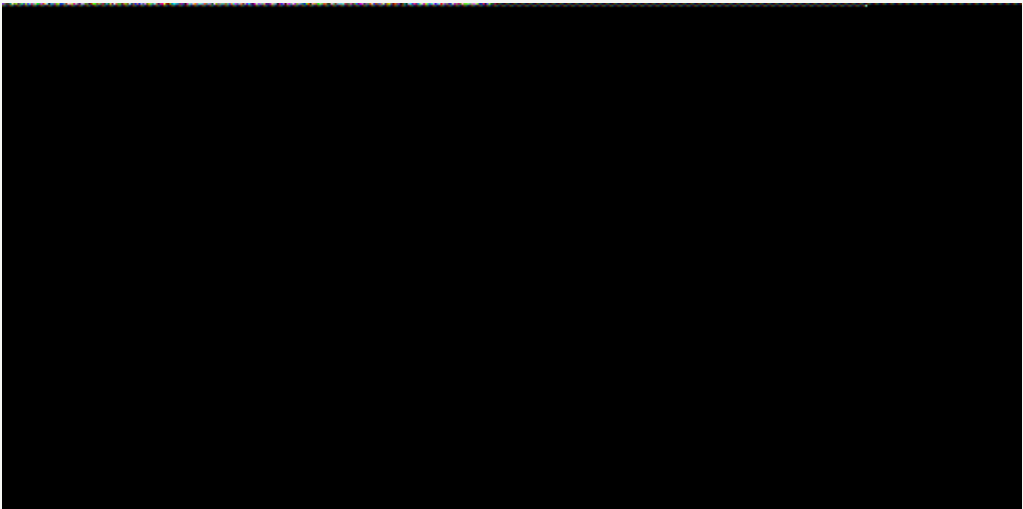
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- ① park entrance stitches in ② grade-separated stitch north of N3rd Ring
- ③ station-hinterland slow modes ④ Gaoliangqiao Xiejie interface
- Interface Plaza · Emergence Bell + the 1733 AI-native interface
- Section line C-C' matches the detailed plan

⑩ Street sections of the three areas (conceptual · dimension chain · plan index)

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▪ Each section: building-to-building full section + dimension chain + floors/heights

▪ Scale references: 1.8m people · bikes · cars · low-speed shuttle · trees & lamps

▪ Dazhongsi C-C': 20m interface plaza + stitching skybridge (5.5m) + Emergence Bell

▪ Conceptual sections (not engineering); dimensions pending redline & heritage data

Conceptual sections: design intent; to be deepened on redline, heritage and vertical data



Character & conceptual height zoning

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▪ Height bands step down along the spine: 0–8 / 18 / 24 / 36 m

▪ Three rules: garden blocks / near-campus street / station front

▪ Protective overlay (heritage + view corridors) + generative rules (low front, high back)

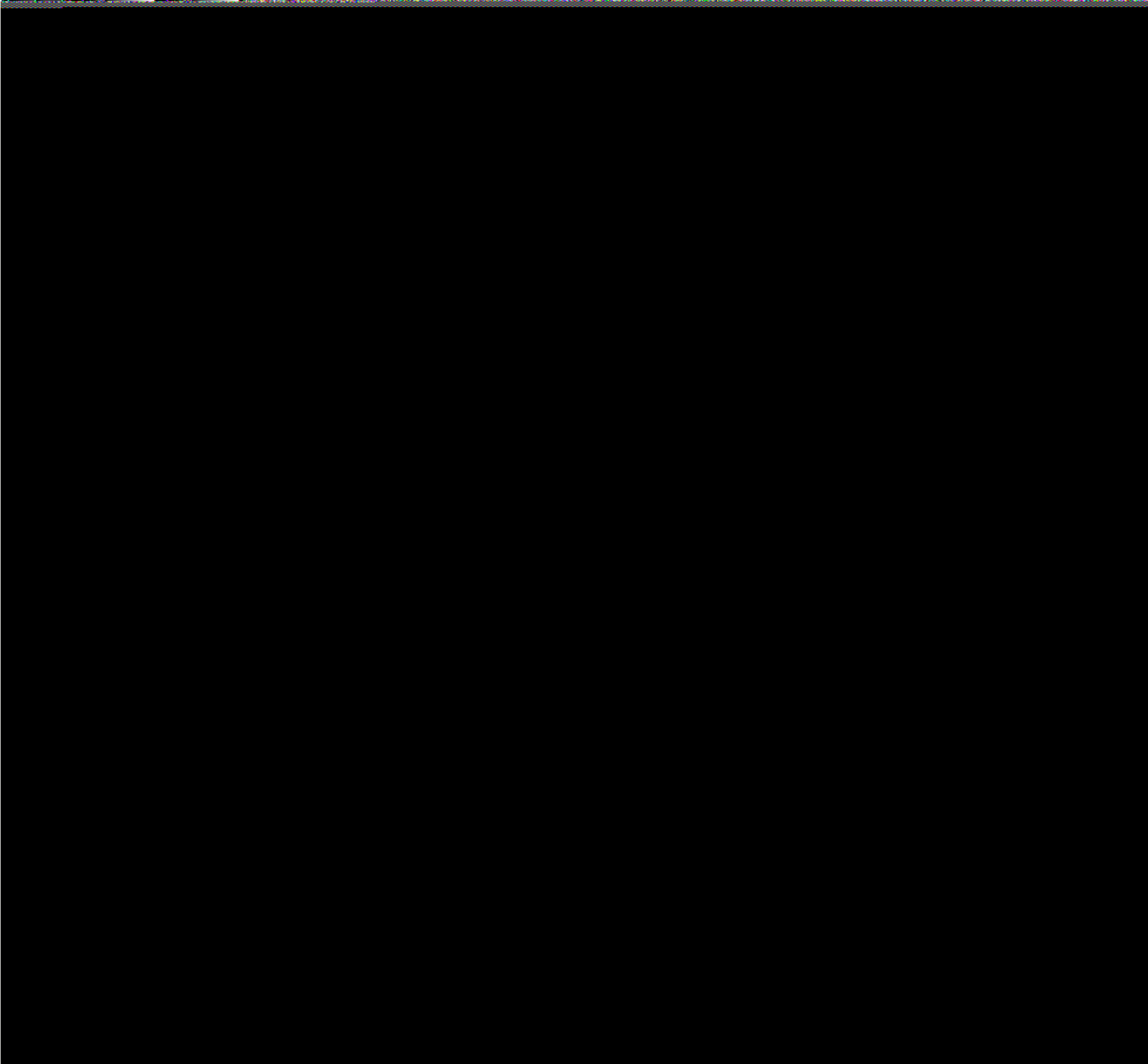
▪ Height values await heritage & regulatory data (not a regulatory drawing)



03

Scenarios & ecosystem: the AI ecology and pilgrimage

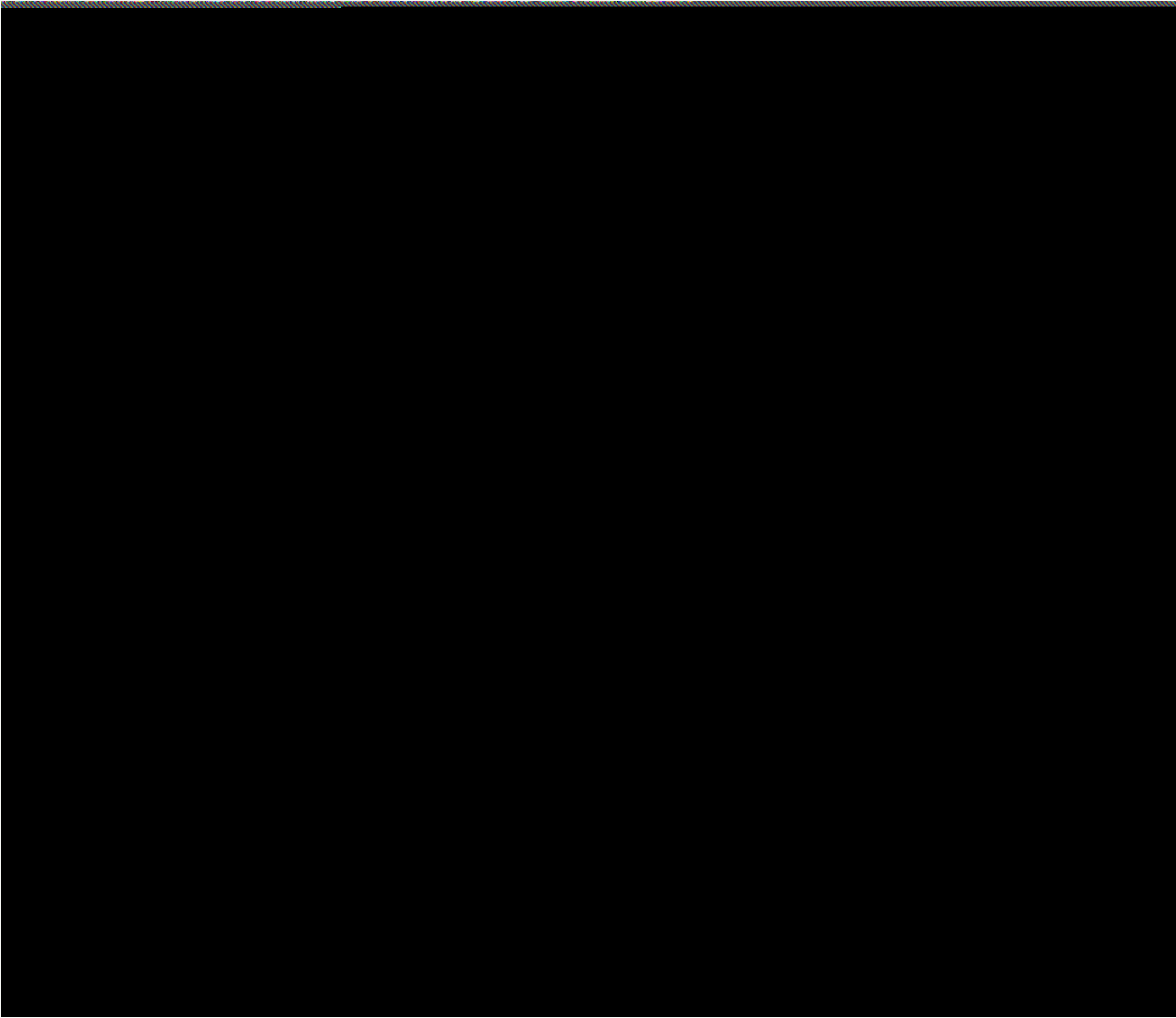
Ecosystem map & seven cases · Ten scenario cards, five personas · Three pilgrimage landmarks · Operations & timeline



- Upstream complete: 1,000+ scientists, 13k developers, 180 OPCs, 32 chips
- Middle layer unmeasured: affordability/conversion/compute → annualized
- Seven cases, one transferable mechanism each (King's Cross / Kendall / Cambridge...)
- Five-party six-force conversion + the annual Phase-Forum report as interface

Operations & the ten-year timeline

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- Annual: dev festival / Phase Forum (λ & H report) / open test seasons
- Attraction funnel: OPC contest → order plaza → desks → incubator → conversion
- Two acceptance gates: $\lambda \geq 40\%$, $H \geq 1.35$ — or stop and review
- Phases: 1–3 stitching+interface / 3–5 Origin / 5–10 Zhongzhiyuan



International benchmark matrix: cases × mechanisms × responses × numbers

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▪ Seven global cases, one transferable mechanism each (no copied images or geometry)

▪ King's Cross × stitch-first / Kendall × conversion circle / Stanford × garden neighbourhoods

▪ Yuehai × dense fabric / Shibuya × consumption / Cambridge × symbiosis / Rothschild × public life

▪ The opportunity: seven mechanisms hold at once on one belt — Jing-Zhang's unique window

Benchmark discipline: mechanisms transferred, numbers computed locally

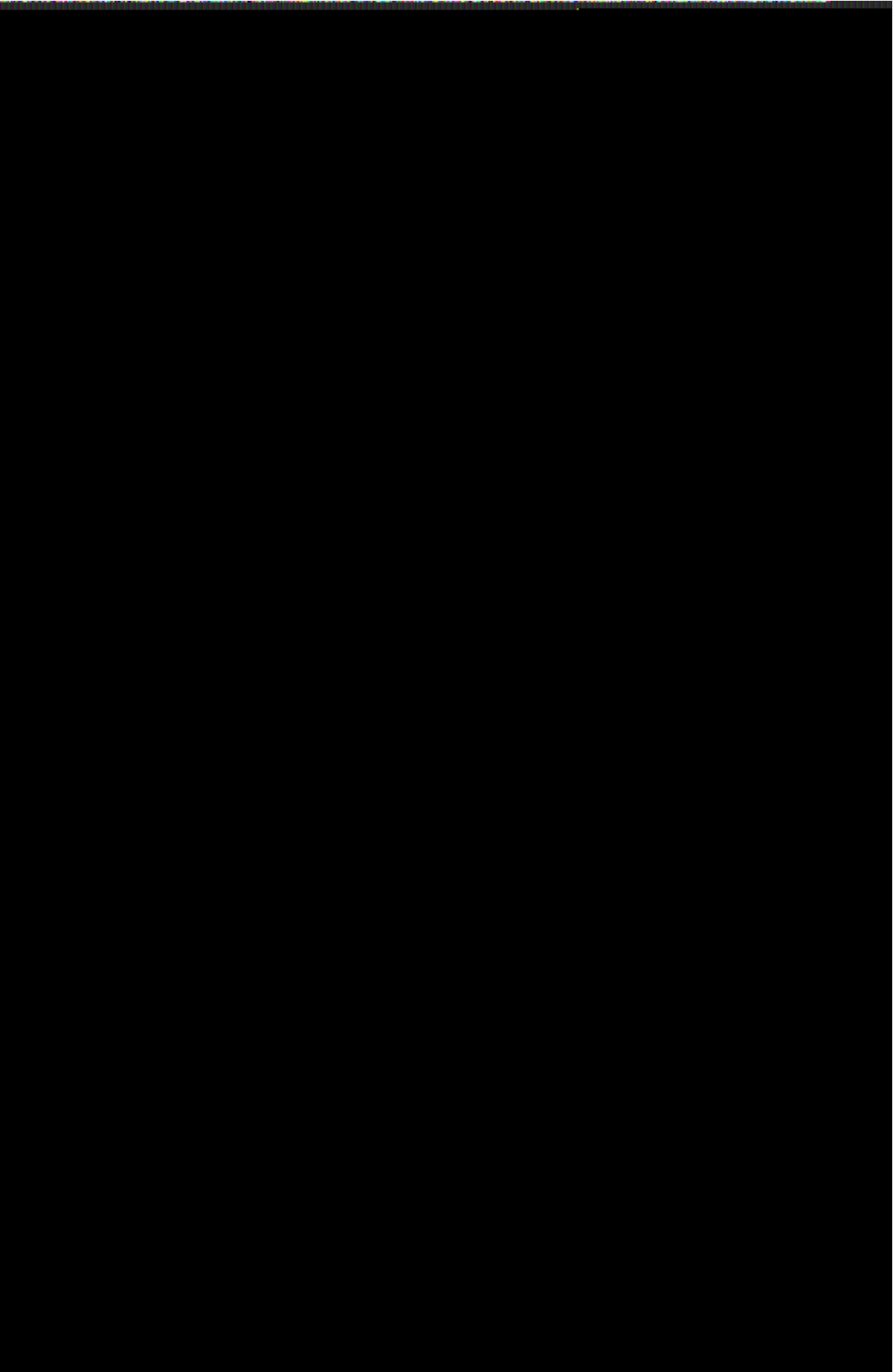
04

System & finance: the Emergent Brain and the investment ledger

Emergent Brain architecture · Phase Control Room · Investment & return · Risk & compliance

▣ The Emergent Brain: the city as a recursive program

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- Three layers: sense → compute → decide, with humans always holding the veto
- Four terminals: citizens / enterprises / government / campus
- The recursive loop is the annual check-up:
read→compute→tune→verify→re-read
- Red lines: no privacy violation, no over-monitoring, human-reviewable

Phase Control Room: the belt's annual check-up

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▪ Four gauges: λ 29.2%→40% target / H 1.556 / D 1.746 / p 83.9%

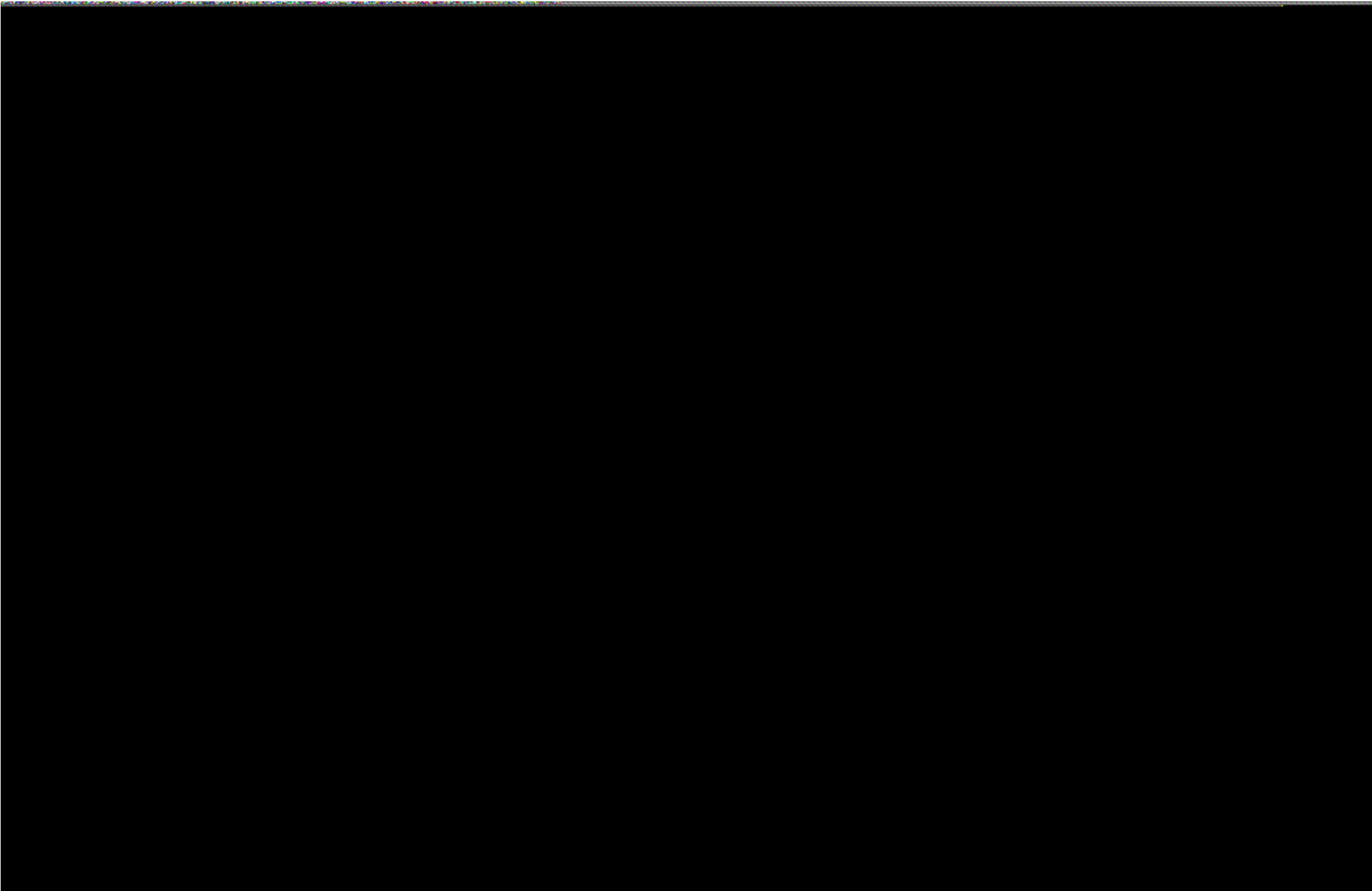
▪ Live panels: job reach 323→375, match 46.2%→61.5%, η +60%

▪ Physically at Wudaokou · Phase Plaza: a public instrument, not surveillance

▪ Aggregate only, zero raw points, zero PII, human veto

Investment & return (conceptual magnitude)

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▪ CNY 4.40bn: stitching 0.66 / space 0.24 / interface 0.54 / apartments 1.56 / test band 0.60 / brain 0.50 / ops 0.30

▪ Three ledgers: rent (IRR≈0.5% alone) + asset (y10 REITs/disposal, blended IRR≈10.6%) + social

▪ Demand: 4,919 work POIs → 100k m² offices; 38.4% balance → 2,000 apartments

▪ Public ledger ≈1.7bn seeks no financial return; market ledger ≈2.7bn runs rent+asset

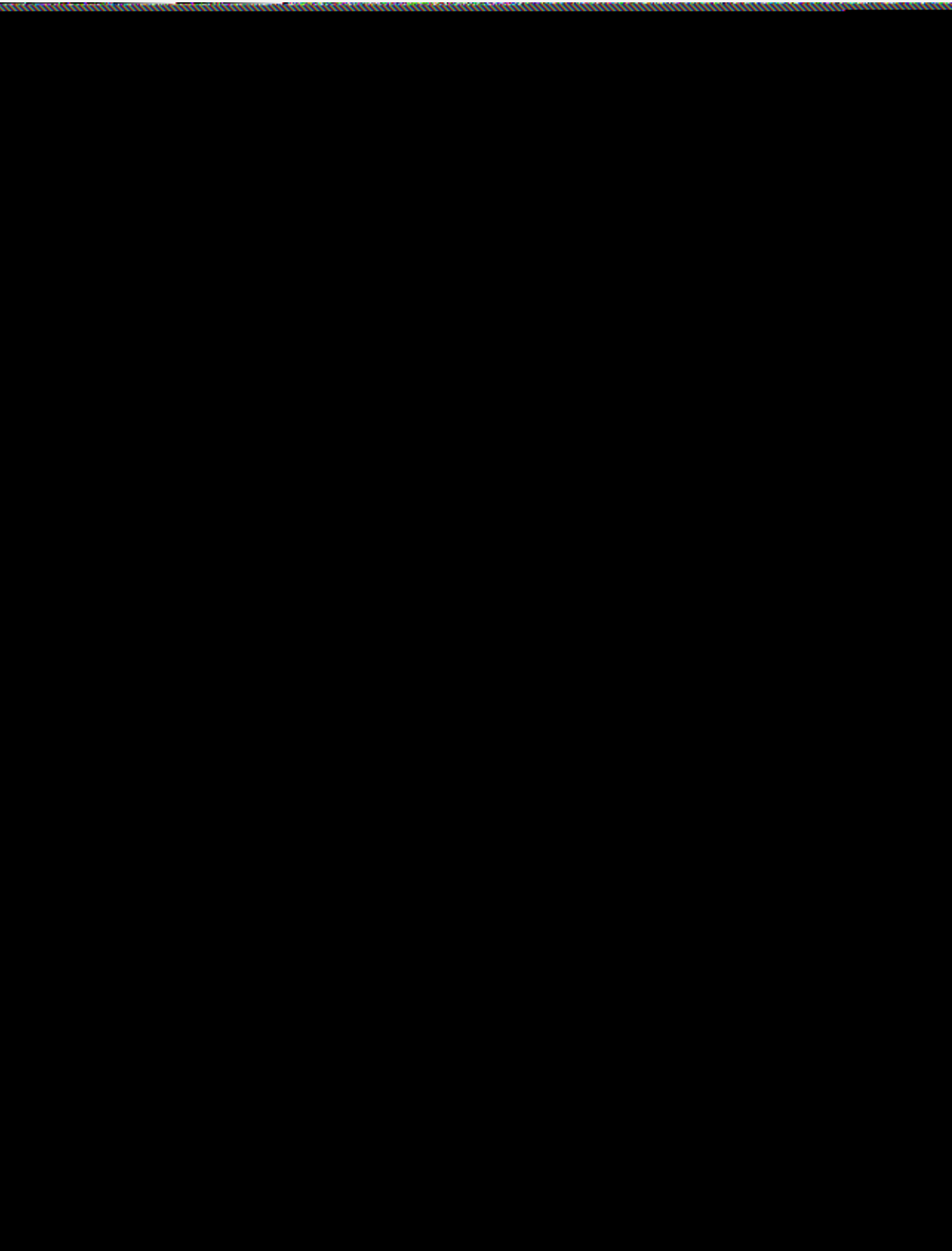
Conceptual; not an investment commitment; rents & occupancy are replaceable assumptions



05

Conclusion: four sentences, each with numbers

Conclusion dashboard · 1909 switch → 2026 belt



▪ ① Re-grow, not rebuild: D=1.746 / H=1.556 / giant 83.9%

▪ ② Move aside what blocks: 7 gaps→7 stitches / 42 cells / three annualized

▪ ③ Falsifiable: jobs +15.9% / match +15.3pct / η +60%

▪ ④ An OS, not a blueprint: four rulers + brain + check-up + exit

Move aside what blocks emergence, and leave the rest for the city to grow itself.

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KUN-SAL Spatial Quant Lab · [GitHub: Anshengdesign](#)

This proposal is an open co-creation suggestion; it does not replace formal planning and constitutes no government-reviewed conclusion; provisional boundaries are dashed only, statutory indicators pending.